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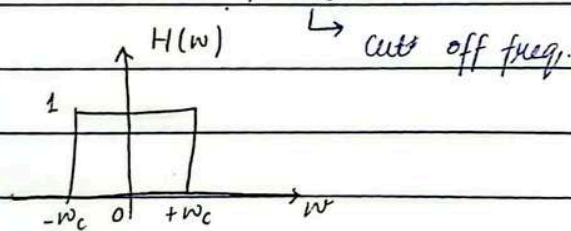
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CHAPTER #10 DIGITAL FILTERS

IDEAL LOW PASS:

The impulse response $h(n)$ of an ideal low pass filter with frequency characteristics is:

$$H(\omega) = \begin{cases} 1 & ; |\omega| \leq \omega_c \\ 0 & ; \omega_c < \omega \leq \pi \end{cases} \quad \therefore -\omega_c < \omega < \omega_c$$



Impulse response: $h(n)$:

$$\text{IDTFT} = \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} H(\omega) e^{j\omega n} d\omega$$

$$h(n) = \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} 1 e^{j\omega n} d\omega$$

$$= \frac{1}{2\pi} \left[\frac{e^{j\omega n}}{jn} \right]_{-\omega_c}^{\omega_c}$$

$$= \frac{1}{2\pi} \left[\frac{e^{j(2\omega_c)n} - e^{-j(2\omega_c)n}}{2jn} \right]$$

$$= \frac{1}{2\pi} \left[\frac{2 \sin(n\omega_c)}{n} \right]$$

$$= \frac{\sin(n\omega_c)}{\pi n}, \quad n \neq 0$$

$$\sin x = \frac{e^{jx} - e^{-jx}}{2j}$$

$$\Rightarrow 2 \sin x = \frac{e^{jx} - e^{-jx}}{j}$$

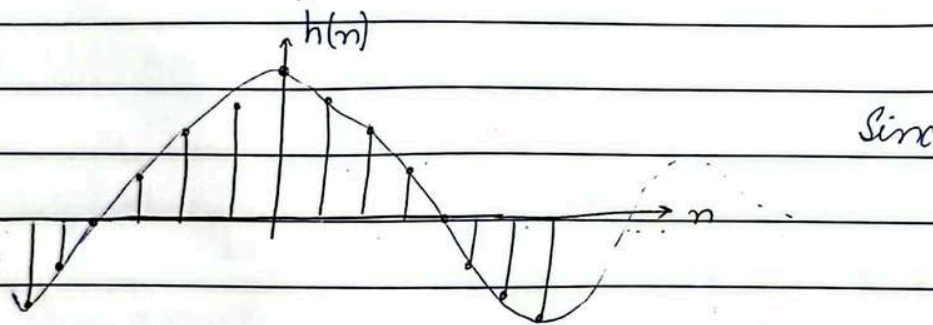
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For $n=0$

$$h(n) = \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} e^{j\omega n} d\omega$$
$$= \frac{\omega_c}{\pi}$$

$$\Rightarrow h(n) = \begin{cases} \frac{\omega_c}{\pi} & ; \quad n=0 \\ \frac{\sin(n\omega_c)}{\pi n} & ; \quad n \neq 0 \end{cases}$$

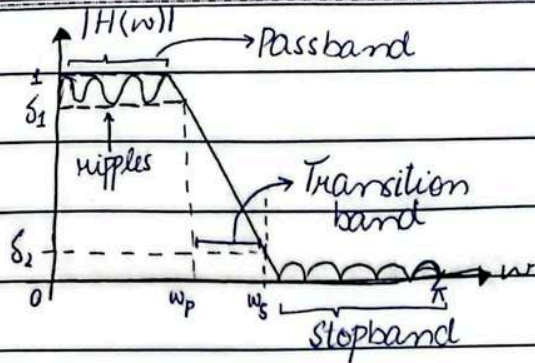


Unit sample response of an ideal low pass filter.

CHARACTERISTICS OF PRACTICAL FREQUENCY SELECTIVE FILTERS: (signal having ∞ samples)

- As we observed that ideal filters are unrealizable for real-time signal processing applications. They are not absolutely necessary in most practical applications. A small amount of ripple is usually ~~to~~ tolerable in passband. Similarly, a small non-zero value or small amount of ripple is also tolerable in stopband.
- The transition of the frequency response from the Passband to stopband defines the transition band.

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 ω_p = Edge of passband ω_s = Edge of stopband

* Width of the passband is called bandwidth of filter

Magnitude characteristics of physically realizable filters

finite impulse response

DESIGN OF LINEAR PHASE FIR FILTERS USING WINDOWS

- In this method, we begin with the desired frequency response specification $H_d(\omega)$ and determine the corresponding unit sample response $h_d(n)$. Indeed, $h_d(n)$ is related to $H_d(\omega)$ by the Fourier Transform relation:

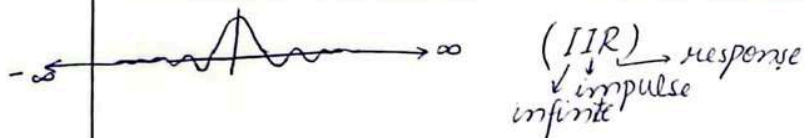
$$H_d(\omega) = \sum_{n=0}^{\infty} h_d(n) e^{-j\omega n}$$

where,

$$h_d(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} H_d(\omega) e^{j\omega n} d\omega \quad (i)$$

- Thus, given $H_d(\omega)$, we can determine $h_d(n)$ by evaluating the integral in eq. (i).
- In general, $h_d(n)$ is infinite in duration and must be truncated at some point, say at $n = M-1$, to yield an FIR filter of length M .
- Truncation of $h_d(n)$ to a length $M-1$ is equivalent to multiplying $h_d(n)$ by a "rectangular window" defined as:

$$w_M(n) = \begin{cases} 1 & ; \quad n = 0, 1, 2, \dots, M-1 \\ 0 & ; \quad \text{otherwise} \end{cases}$$



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Thus, the unit sample response of the FIR filter becomes:

$$h(n) = \sum h_d(n) w(n)$$

$$h(n) = \begin{cases} h_d(n) & ; n=0, 1, 2, \dots, M-1 \\ 0 & ; \text{otherwise} \end{cases}$$

17/4 Design of Linear-phase FIR filters using windows

$$W(\omega) = \sum_{n=0}^{M-1} e^{-j\omega n}$$

Apply the geometric series formula:

$$\sum_{n=0}^{M-1} r^n = \frac{1-r^M}{1-r} \quad ; \text{ for } r \neq 1$$

$$= \frac{1-e^{-j\omega M}}{1-e^{-j\omega}} \Rightarrow r = e^{-j\omega}$$

Factor out $e^{-j\omega M/2}$ from numerator & $e^{-j\omega/2}$ from den.

$$= \frac{e^{-j\omega M/2} (e^{j\omega M/2} - e^{-j\omega M/2})}{e^{-j\omega/2} (e^{j\omega/2} - e^{-j\omega/2})}$$

$\Rightarrow$ Use Euler's identity $\sin(x) = \frac{e^{jx} - e^{-jx}}{2j}$

$$= \frac{e^{-j\omega M/2} \times 2j \sin(\omega M/2)}{e^{-j\omega/2} \times 2j \sin(\omega/2)}$$

$$W(\omega) = e^{-j\omega(M-1)/2} \cdot \frac{\sin(\omega M/2)}{\sin(\omega/2)}$$

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Magnitude (dB):

$$|W(\omega)| = \frac{|\sin(\omega M/2)|}{|\sin(\omega/2)|} \quad ; \quad \pi \leq \omega \leq \pi$$

We know,

$$|e^{-j\omega(M-1)/2}| = 1$$

$$|e^{j\theta}| = \sqrt{\cos^2 \theta + \sin^2 \theta}$$

Phase:

$$W(\omega) = \begin{cases} -\omega \left(\frac{M-1}{2} \right) & ; \text{ when } \sin(\omega M/2) > 0 \\ -\omega \left(\frac{M-1}{2} \right) + \pi & ; \text{ when } \sin(\omega M/2) < 0 \end{cases}$$

$$W(\omega) = |W(\omega)| e^{j\theta}$$

$$\theta = -\omega \left(\frac{M-1}{2} \right)$$

EXAMPLE # 8.3.4

$$H_a(s) = \frac{s + 0.1}{(s + 0.1)^2 + 16} \quad (i)$$

$$(s + 0.1)^2 + 16$$

$$= s + 0.1$$

$$(s + 0.1)^2 + (4)^2$$

$$\Omega_n = 4 \text{ (resonant frequency)}$$

We know,

$$\Omega = \frac{2}{T} \tan(\omega/2)$$

$$\Rightarrow \overset{=4}{\Omega_n} = \frac{2}{T} \tan(\omega/2) \Rightarrow T \times 4 = 2 \tan(\pi/4)$$

$$T = 1/2 \times 1 = 1/2$$

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We know,
$$S = \frac{2}{T} \left(\frac{1-z^{-1}}{1+z^{-1}} \right) = 4 \left(\frac{1-z^{-1}}{1+z^{-1}} \right) \quad (iii) \quad \because T = 1/2$$

Put (iii) in (i)

$$H(z) = \frac{4 \left(\frac{1-z^{-1}}{1+z^{-1}} \right) + 0.1}{\left(4 \times \frac{1-z^{-1}}{1+z^{-1}} + 0.1 \right)^2 + 16}$$

Numerator:
$$= \frac{4(1-z^{-1}) + 0.1(1+z^{-1})}{1+z^{-1}} = \frac{4.1 - 3.9z^{-1}}{1+z^{-1}} \quad (iii)$$

Denominator:
$$= \left(\frac{4.1 - 3.9z^{-1}}{1+z^{-1}} \right)^2 + 16 = \frac{(4.1 - 3.9z^{-1})^2 + 16(1+z^{-1})^2}{(1+z^{-1})^2}$$

$$= \frac{16.81 + 0.21z^{-2} - 31.98z^{-1} + 16 + 32z^{-1} + 16z^{-2}}{(1+z^{-1})^2}$$

$$= \frac{32.81 + 0.02z^{-1} + 32.21z^{-2}}{(1+z^{-1})^2}$$

Combine Num & Den.

$$= \frac{4.1 - 3.9z^{-1}}{1+z^{-1}} \times \frac{(1+z^{-1})^2}{32.81 + 0.02z^{-1} + 32.21z^{-2}}$$

$$= \frac{4.1 + 0.2z^{-1} - 3.9z^{-2}}{32.81 + 0.02z^{-1} + 32.21z^{-2}}$$

÷ Num & den by 32.81 to make const. term in den = 1

$$H(z) = \frac{0.125 + 0.0061z^{-1} - 0.1189z^{-2}}{1 + 0.0006z^{-1} + 0.975z^{-2}} = \frac{0.125 + 0.0061z^{-1} - 0.1189z^{-2}}{1 + 0.975z^{-2}}$$

extremely
small $\Rightarrow$
can be
ignored

* Multiply num & den by z^2 to remove -ve power.

$$H(z) = \frac{0.125z^2 + 0.0061z - 0.1189}{z^2 + 0.975}$$

- The filter has poles at $P_{1,2} = 0.981e^{\pm j\pi/2}$
- The filter has zeros at $Z_{1,2} = -1, 0.95$

8.3.5

CHARACTERISTICS OF COMMONLY USED ANALOG

FILTERS:

- IIR digital filters can easily be obtained by beginning with an analog filter and then using a mapping to transform the s -plane into z -plane.

Thus, the design of a digital filter is reduced to designing an appropriate analog filter and then performing the $\downarrow$ conversion from $H(s)$ to $H(z)$ plane.

→ BUTTERWORTH FILTER:

Low pass butterworth filters are all-pole filters characterized by the following magnitude-squared frequency response:

$$|H(\Omega)|^2 = \frac{1}{1 + (\Omega/\Omega_c)^{2N}}$$

$\therefore N$ is the order of the filter, Ω_c is the -3 dB cut off frequency.

Since, $H(s)H(-s)$ evaluated at $s = j\Omega$ is simply equal to the $|H(\Omega)|^2$, it follows that:

$$|H(\Omega)|^2 = H(s)H(-s) = \frac{1}{1 + (-s^2/\Omega_c^2)^N} \quad (i)$$

From eq. (i), we find that:

$$\frac{-s^2}{\Omega_c^2} = (-1)^{1/N} = e^{j(2k+1)\pi/N} \quad ; k = 0, 1, 2, \dots, N-1$$

& hence,

$$s_k = \Omega_c e^{j\pi/2} e^{j(2k+1)\pi/2N} \quad ; k = 0, 1, 2, \dots, N-1$$

$$\Omega = \omega \text{ omega}$$

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$$|H(\Omega)|^2 = H(s) H(-s) = \frac{1}{1 + (\Omega/\Omega_c)^{2N}}$$

$$= \frac{1}{1 + (\sqrt{\Omega^2}/\Omega_c)^{2N}}$$

$$= \frac{1}{1 + \left(\frac{\sqrt{-s^2}}{\Omega_c}\right)^{2N}}$$

$$\begin{aligned} \sqrt{\Omega^2} &= \Omega \\ \therefore -s^2 &= \Omega^2 \end{aligned}$$

$$\text{At } s = j\Omega \Rightarrow s^2 = j^2 \Omega^2 = -\Omega^2 \Rightarrow -s^2 = \Omega^2$$

$$|H(\Omega)|^2 = H(s) H(-s) = \frac{1}{1 + (-s^2/\Omega_c^2)^N}$$

Find eq. for poles:

$$1 + \left(\frac{-s^2}{\Omega_c^2}\right)^N = 0$$

$$\left(\frac{-s^2}{\Omega_c^2}\right)^N = -1$$

$$\Rightarrow \frac{-s^2}{\Omega_c^2} = (-1)^{1/N} \quad (ii)$$

We know:

$$-1 = e^{j\pi} \quad (iii) \quad \therefore e^{j\pi} = \cos \pi + j \sin \pi = -1$$

$$e^{j\pi} = e^{j(\pi + 2\pi k)}$$

$$\therefore e^{j\pi} = e^{j(2k+1)\pi}$$

(Since adding $2\pi k$ does not change any value)

$\Rightarrow$ (iii) in (ii)

$$\frac{-s^2}{\Omega_c^2} = e^{j\pi/N} = e^{j(2k+1)\pi/N} \quad ; k = 0, 1, 2, \dots, N-1$$

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$$-s^2 = -\Omega_c^2 e^{j(2k+1)\pi/N}$$

$$s^2 = \Omega_c^2 (-1) e^{j(2k+1)\pi/N}$$

$$(-1) = e^{j\pi}$$

$$s_k = \sqrt{\Omega_c^2} \sqrt{e^{j\pi}} \sqrt{e^{j(2k+1)\pi/N}}$$

$$\sqrt{\dots} = (\dots)^{1/2}$$

$$* \quad s_k = \Omega_c e^{j\pi/2} e^{j(2k+1)\pi/2N}$$

$$\text{eq (i)} \rightarrow |H(\Omega)|^2 = \frac{1}{1 + \left(\frac{-s^2}{\Omega_c^2}\right)^N}$$

The order of the filter can be obtained from eq (i)
We want "N" to be have attenuation δ^2 at a specified frequency Ω_s . Thus at $\Omega = \Omega_s$,

$$|H(\Omega_s)|^2 = \frac{1}{1 + \left(\frac{-\Omega_s^2}{\Omega_c^2}\right)^{2N}} = \delta^2$$

$$\Rightarrow \frac{1}{\delta^2} = 1 + \left(\frac{\Omega_s}{\Omega_c}\right)^{2N}$$

$$\Rightarrow \left(\frac{\Omega_s}{\Omega_c}\right)^{2N} = \frac{1}{\delta^2} - 1$$

Take natural log

$$2N \log\left(\frac{\Omega_s}{\Omega_c}\right) = \log\left(\frac{1}{\delta^2} - 1\right)$$

$$* \quad \Rightarrow N = \frac{\log\left(\frac{1}{\delta^2} - 1\right)}{2 \log\left(\frac{\Omega_s}{\Omega_c}\right)} \quad (\text{in}) \quad (\text{To find order of filter})$$

$$\text{At } \Omega = 0 \Rightarrow |H(\Omega)|^2 = 1$$

$$\text{At } \Omega = \Omega_c \Rightarrow |H(\Omega_c)|^2 = 1/2$$

$$\text{At } \Omega \gg \Omega_c \Rightarrow \left(\frac{\Omega_c}{\Omega}\right)^{2N} \approx |H(\Omega)|^2$$

approaches zero

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EXAMPLE # 8.3.6

Determine the order and poles of a lowpass butterworth filter that has a -3dB bandwidth of 500Hz, and an attenuation of 40dB at 1000Hz.

Sol.

$$\begin{aligned}
 & \downarrow \begin{aligned} & \delta^2 = 0.01 \Rightarrow \delta^2 = 10^{-4} \\ & \downarrow 1/\delta^2 = 100 \end{aligned} \\
 & \left[\begin{aligned} & \delta^2 = \frac{1}{1 + (\Omega_s/\Omega_c)^{2N}} \\ & = \frac{1}{1 + (2)^{2N}} \end{aligned} \right] \quad \left[\begin{aligned} & = \frac{1}{1 + (1000/500)^{2N}} \\ & = \frac{1}{1 + 4^N} \end{aligned} \right] \quad \begin{aligned} & \delta^2 \Rightarrow 20 \log(1/\delta) = \text{dB} \\ & 20 \log(1/\delta^2) = 40 \\ & \delta = 0.01 \end{aligned} \\
 & N = \frac{\log[(1/\delta^2) - 1]}{2 \log(1000/500)} = \frac{\log[(1/0.01^2) - 1]}{2 \log(2)} \\
 & N = 6.64
 \end{aligned}$$

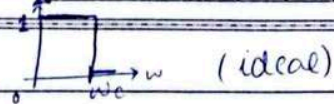
$$S_k = \Omega_c e^{j\pi/2} e^{j(2k+1)\pi/2N} \quad ; \quad k = 0, 1, 2, \dots, N-1$$

$$\text{Poles: } S_k = 1000 e^{j\pi/2} e^{j(2k+1)\pi/14} \quad ; \quad k = 0, 1, 2, 3, 4, 5, 6$$

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BUTTERWORTH FILTER



$$\text{Magnitude of frequency response} = |H(\Omega)|^2 = \frac{1}{1 + (\Omega/\Omega_c)^{2N}}$$

$$= \frac{1}{1 + (-s^2/\Omega_c^2)^N} \quad \text{where } -s^2 = \Omega^2$$

$\Omega_c = -3\text{dB/cut off frequency}$

$$S_k = \Omega_c e^{j\pi/2} e^{j(2k+1)\pi/2N} \quad \text{for } k=0, 1, 2, \dots, N-1$$

$$N = \frac{\log[(1/s^2) - 1]}{2 \log(\Omega_s/\Omega_c)}$$

$$|H(\Omega)|^2 = \frac{1}{1 + (\Omega/\Omega_c)^{2N}}$$

→ At $\Omega = 0$ (DC frequency)

$$|H(\Omega)|^2 = \frac{1}{1+0} = 1$$

→ At $\Omega = \Omega_c$

$$|H(\Omega_c)|^2 = \frac{1}{1 + (\Omega_c/\Omega_c)^{2N}} = \frac{1}{2}$$

→ At $\Omega \gg \Omega_c$, let $\Omega = 1000$, $\Omega_c = 1$

$$|H(\Omega)|^2 = \frac{1}{1 + (1000/1)^{2N}} \approx \frac{1}{(1000/1)^{2N}}$$

→ insignificant ←

$$\approx \left(\frac{1}{1000}\right)^{2N} \approx \left(\frac{\Omega_c}{\Omega}\right)^{2N}$$

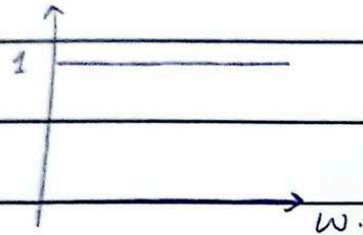
$= 0.0010$

Hence, $|H(\Omega)|^2$ approaches zero as $\Omega \gg \Omega_c$. So for higher frequencies than Ω_c , $|H(\Omega)|^2$ behaves like "stop band".

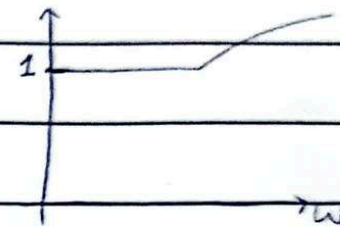
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* if $\Omega \gg \Omega_c \Rightarrow H(\Omega) \uparrow$



OR



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CHAPTER #8

POLES OF BUTTERWORTH (BW) FILTERS:

$$S_k = \Omega_c e^{j\pi/2} e^{j(2k+1)\pi/2N}$$

$$|S_k| = \Omega_c / e^{j\pi/2} e^{j(2k+1)\pi/2N} \quad ; \quad k = 0, 1, 2, \dots, N-1$$

$$|S_k| = \Omega_c \quad \because |e^{j\pi/2} e^{j(2k+1)\pi/2N}| = 1$$

Poles of Bandwidth filter lie on a circle of radius Ω_c .

Let,

$$N = 4$$

$$k = 0, 1, 2, 3$$

$$k = 0$$

$$S_0 = \Omega_c e^{j\pi/2} e^{j(1)\pi/2 \times 4}$$

$$= \Omega_c e^{j\pi/2} e^{j\pi/8}$$

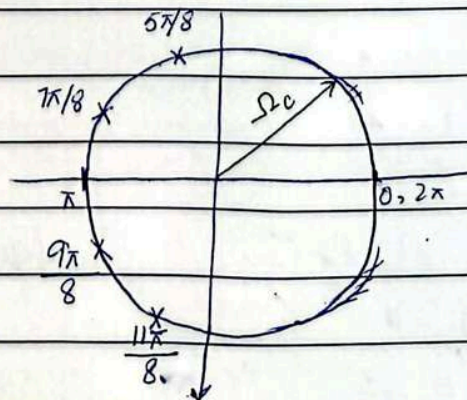
$$S_0 = 1 e^{j(\pi/2 + \pi/8)} = e^{j\theta}$$

$$\theta_0 = \frac{\pi}{2} + \frac{\pi}{8} = \frac{5\pi}{8}$$

$$\theta_1 = \frac{7\pi}{8}$$

$$\theta_2 = \frac{9\pi}{8}$$

$$\theta_3 = \frac{11\pi}{8}$$



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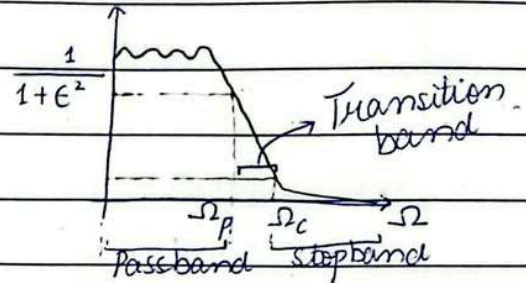
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* frequency response $\Rightarrow$

$$|H(\Omega)|^2 = \frac{1}{1 + \epsilon^2 T_N^2(\Omega/\Omega_p)}$$

$\downarrow x$



ϵ = parameter related to the passband ripple in the passband.

$T_N(x)$ = N^{th} order polynomial defined as:

*
$$T_N(x) = \begin{cases} \cos(N \cos^{-1} x) & ; |x| \leq 1 \\ \cosh(N \cosh^{-1} x) & ; |x| > 1 \end{cases}$$

RECURSION EQUATION:

$$T_{N+1}(x) = 2x T_N(x) - T_{N-1}(x)$$

$N = 1, 2, \dots$

* $\frac{\Omega}{\Omega_p} = x$

where,

$$T_0(x) = 1$$

$$T_1(x) = x$$

$$T_2(x) = 2x^2 - 1$$

$$T_{1+1}(x) = 2x T_1(x) - T_0(x) = 2x(x) - 1$$

$$T_3(x) = 4x^3 - 3x$$

For 'N' odd

Let, $N = 1$

$$T_1(x) = x$$

for,

$$x = \Omega = 0$$

(given frequency)

$$T_1(x) = 0$$

$$|H(0)|^2 = \frac{1}{1 + \epsilon^2 T_1^2(0)}$$

$$T_1(0) = 0$$

$$1 + \epsilon^2 T_1^2(0)$$

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$|H(0)|^2 = 1$ At $\Omega = 0$, gain is maximum = 1 & attenuation is zero.

For 'N' even:

Let, $N=0$

$$T_0(x) = 1$$

Let, $\Omega = x = 0$

(given frequency)

$$T_0(0) = 1$$

$$|H(0)|^2 = 1$$

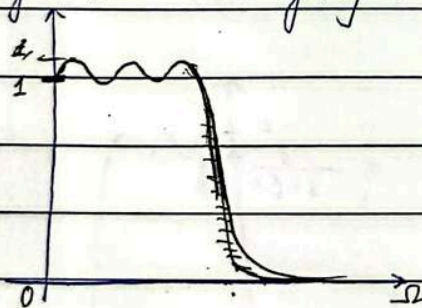
$$1 + \epsilon^2 T_0^2(0)$$

$$= 1$$

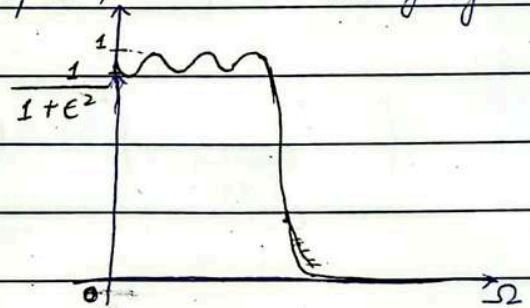
At $\Omega = 0$, gain is slightly less than one, depending on ϵ .

$$1 + \epsilon^2$$

for chebyshev * By ~~also~~ changing the value of ϵ , we can vary gain.



for odd 'N'



for even 'N'

For 'N' odd

At $\Omega = \Omega_p$

$$|H(\Omega)|^2 = 1$$

$$1 + \epsilon^2 T_N^2(\Omega/\Omega_p)$$

$$= 1$$

$$1 + \epsilon^2 T_N^2(\Omega_p/\Omega_p)$$

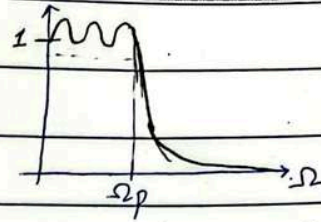
$$= 1$$

$$1 + \epsilon^2 T_N^2$$

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$$|H(\Omega)|^2 = \frac{1}{1 + \epsilon^2}$$



For 'N' even:

$$\text{At } \Omega = \Omega_p$$

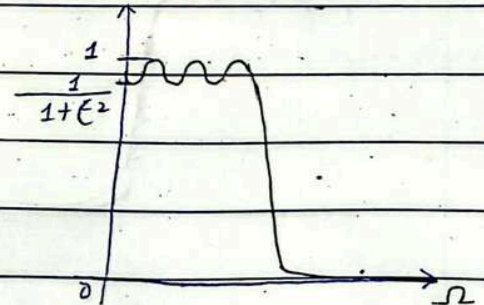
$$|H(\Omega)|^2 = \frac{1}{1 + \epsilon^2 T_N^2(0)}$$

$$|H(\Omega_p)|^2 = \frac{1}{1 + \epsilon^2}$$

Hence, we can write,

$$H(\Omega) = \frac{1}{\sqrt{1 + \epsilon^2}} \approx 1 - \delta_1$$

$$\left(\frac{1}{\sqrt{1 + \epsilon^2}} \right)^2 = (1 - \delta_1)^2$$



$$\frac{1}{1 + \epsilon^2} = (1 - \delta_1)^2$$

$$\epsilon^2 = \frac{1}{(1 - \delta_1)^2} - 1$$

δ_1 = value of the passband ripple.

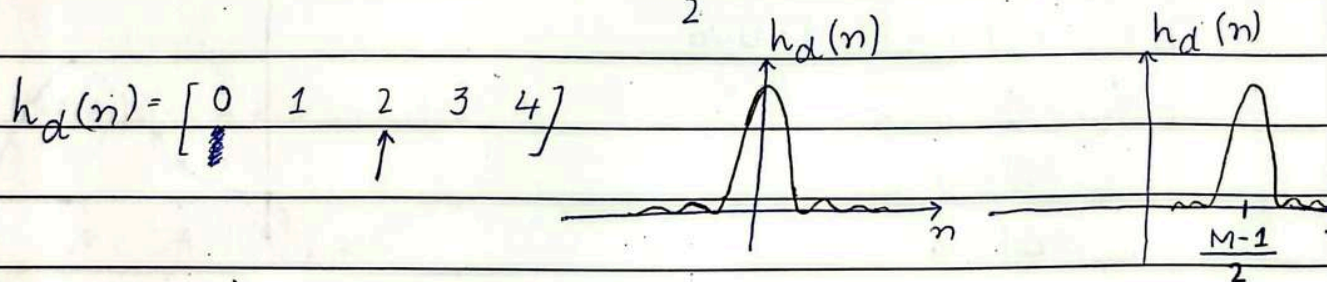
DESIGN OF FIR FILTERS WITH WINDOWS

- In real life applications, we initiate designing filter in frequency domain. For example, we initiate from ω_c/Ω_c , Ω_c , etc.

- Second, a designed filter should be causal and its length should be 'M' from $n=0$ to $n=M-1$

- To hold the symmetry within desired impulse response $h_d(n)$ we want the filter to center at middle of M length filter. Hence, we shift (the center to $(\frac{M-1}{2})$ by highest value

multiplying $H_d(j\omega)$ with $e^{-j\omega(\frac{M-1}{2})}$. Hence, if $M=5$, then the center will be at $\frac{5-1}{2} = 2$ i.e.



DESIGN A LPF OF LENGTH $M=5$, $\omega_c = \pi/2$

CASE-I: WITHOUT MULTIPLYING $\exp(-j\omega((M-1)/2))$ WITH $H_d(\omega)$:

$$H(\omega) = \begin{cases} 1 & ; |\omega| \leq \omega_c \\ 0 & ; \text{otherwise} \end{cases}$$

$$h(n) = \begin{cases} 0.5 & ; n=0 \\ \frac{0.5 \sin(\pi n/2)}{\pi n} & ; n \neq 0 \end{cases}$$

$$h_d(n) = h(n) \cdot w(n)$$

↓
window

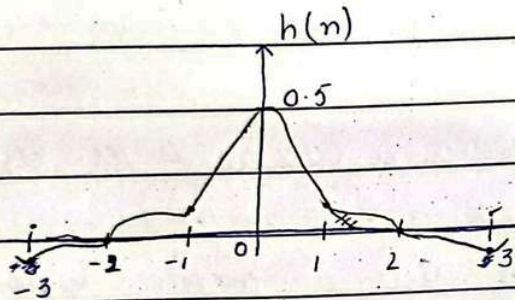
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$$h(0) = 0.5, \quad h(-1) = 0.16, \quad h(1) = 0.16$$

$$h(2) = 0, \quad h(-2) = 0$$

$$h(-3) = h(3) = -0.05$$



CASE-II: WITH MULTIPLYING $\exp\left(-j\omega \left(\frac{M-1}{2}\right)\right)$ WITH $H_d(\omega)$:

As filter will center at 2. So, compute for $n = 0, 1, 2, 3, 4$

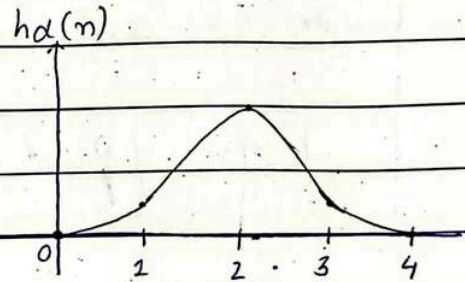
$$h_d(0) = h(n-2) = 0$$

$$h_d(1) = h(-1) = 0.16$$

$$h_d(2) = 0.5$$

$$h_d(3) = 0.16$$

$$h_d(4) = 0$$



Now, $h_d(n)$ is from 0 to 4, which is practically realizable.

Q8. Design a FIR digital filter approximating the ideal frequency response

$$H_d(\omega) = \begin{cases} 1 & ; \quad |\omega| \leq \pi/6 \\ 0 & ; \quad \pi/6 < |\omega| \leq \pi \end{cases}$$

a) Determine the coefficients of 25 tap filter based on the window method with a rectangular window.

Date: _____

Q8.1 Sol $H_d(w) = \begin{cases} 1 e^{-j12w} & 0 \leq w \leq \pi/6 \\ 0 & \text{otherwise} \end{cases}$

$$h_d(n) = \frac{1}{2\pi} \int_{-\pi/6}^{\pi/6} e^{-j12w} \cdot e^{jwn} dw$$

$$= \frac{1}{2\pi} \int_{-\pi/6}^{\pi/6} e^{jw(n-12)} dw$$

$$= \frac{1}{2\pi} \times \frac{e^{jw(n-12)}}{j(n-12)} \Big|_{-\pi/6}^{\pi/6}$$

$$= \frac{1}{2\pi} \left[\frac{e^{j\pi/6(n-12)} - e^{-j\pi/6(n-12)}}{j(n-12)} \right]$$

Multiply & divide by '2'

$$= \frac{2}{2\pi(n-12)} \left[\frac{e^{j\pi/6(n-12)} - e^{-j\pi/6(n-12)}}{2j} \right]$$

$$\neq h_d(n) = \frac{\sin(\pi/6 \times (n-12))}{\pi(n-12)}$$

$$h(n) = h_d(n) \times w(n)$$

where,

$w(n)$ is a rectangular window.

Window of length - 25